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Development Changes in PD-Phase
产品开发更改

Edition 06

Doc. Approval: RBCD/EIS

Effective date: 20250714

DC No. / 开发更改编号: PDECR_0053

Date /日期: 20251105

Step 1: Change request & Basic information

Customer project Name/ 客户项目名称:	57cc e-compressor	MCR No. / MCR号:	NA
Affect Product No. / 产品		Component No./ 零部件号:	
57cc e-compressor F01Z 7P0 013-04 CNHTC 57cc e-compressor F01Z 7P0 00H-01 XCMG 57cc e-compressor F01Z 7P0 012-04 Geely FD 57cc e-compressor F01Z 7P0 012 Geely SOP 57cc e-compressor F01Z 7P0 014-02 Qingling H47 45cc e-compressor F01Z 7P0 01E-01 Qingling 4.5T BEV		**Offerdrawing:** F01Z7D0013-01 F01Z7D000H-01 F01Z7D0012-01 F01Z7D0012 F01Z7D0014-01 F01Z7D001E-01	**EOL function PMB:** F01Z7S0003 F01Z7S0003-01
Sample type / 样品类型:	☐ A ☐ B ☐ C ☑ FD	Initiator/发起人:	ZHU Qiangmin(RBCD/EIS3)

Reason of changes / 更改理由:

1.改善拔出时保护帽冲击力。
2.PMB符合生产实际。

Customer request / 客户要求 ☐	Supplier request / 供应商请求 ☐	Design optimization / 设计优化 ☐	
Correction / 更正(错误) ☐	Process / 工艺 ☐	Other / 其它 ☑	

Step 2: Change proposal & Change discription

Before Change | After Change

1. 更改Offerdrawing中关于压缩机出厂时内部充氮压力，从0.15MPaA~0.25MPaA更改为0.13MPaA~0.19MPaA.

Before Change:

When the compressor leaves the factory, it is internally sealed with dry nitrogen gas of 0.15 MPaA~0.25 MPaA, and when it is switched on, it is discovered that the compressor has no positive pressure, and it should be notified to RB's quality Department for confirmation and use

压缩机出厂时内部封入0.15MPaA~0.25MPaA的干燥氮气，开启使用时发现压缩机没有正压力，应通知博世公司质量部门确认后使用

After removing the inlet and outlet protective caps, it should be

After Change:

When the compressor leaves the factory, it is internally sealed with dry nitrogen gas of 0.13 MPaA~0.19 MPaA, and when it is switched on, it is discovered that the compressor has no positive pressure, and it should be notified to RB's quality Department for confirmation and use 4

压缩机出厂时内部封入0.13MPaA~0.19MPaA的干燥氮气，开启使用时发现压缩机没有正压力，应通知博世公司质量部门确认后使用

After removing the inlet and outlet protective caps, it should be

2.更改功能检测规范3.1.4测试点中关于进口温度变更，从24.0±3°C改为25.0±6°C。

Before Change:

3.1.4 Test points/测试点

Test points 测试点	Setting value			Result specification		
	Set Speed 设定转速 [rpm]	Inlet pressure 进口压力 [barg]	Inlet temperature 进口温度 [ºC]	Outlet pressure 出口压力 [barg]	DC Power current 直流电流 [A]	Rise time 升压时间
1	(1000±150)	(1.0±0.2)	(24.0±3)	(8.3±1.5)	(0.9±0.2)	Record

After Change:

3.1.4 Test points/测试点

Test points 测试点	Setting value			Result specification		
	Set Speed 设定转速 [rpm]	Inlet pressure 进口压力 [barg]	Inlet temperature 进口温度 [ºC]	Outlet pressure 出口压力 [barg]	DC Power current 直流电流 [A]	Rise time 升压时间
1	(1000±150)	(1.0±0.2)	(25.0±6)	(8.3±1.5)	(0.9±0.2)	Record

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Date /日期: 20251105

Step 3.1 Impact analysis / 影响分析

1. Function&Performance will be influenced? / 产品功能性能影响? ☐ yes/是 ☑ no/否

If Yes, product impaction evaluation? 如果是，产品影响评估?（如果是初次样件放行，空白处请说明样件目的）		Testing /签名 朱强敏
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?	不需要	

2.1 Interface&boundary to customer will be influenced? / 接口&边界是否影响客户? ☑ yes/是 ☐ no/否

Please explain the impact and whether it has been confirmed by the customer?请说明影响和是否得到客户的确认?	offerdrawing充氮压力变更需告知客户;	Customer PM /签名

2.2 Interface&boundary to internal will be influenced? / 接口&边界是否影响内部? ☑ yes/是 ☐ no/否

If Yes, internal impaction evaluation? 如果是，内部影响是否评估?（例如：测试接头，噪音测试夹具等）	1.测试线需要更改设定; 2.更改不同压力下保护帽弹出状态试验，见附件。	Testing/签名 朱强敏

3. Mechanical strength&Durability will be influenced? / 产品可靠性、鲁棒性影响？ ☐ yes/是 ☑ no/否

If Yes, How to evaluate? Plan? 如果是，请说明影响和如何评估,计划? (例如:仿真分析等)	NA	Validation/签名
Whether need new sample or fixture to test? Plan? 是否需要样件以及定制/改制夹具进行验证? 计划?		

4. Others component will be influenced? / 影响其它零部件? 不影响

If yes, Which components will change in parallel? 如果影响，哪些零部件也需要同时变更?

5. Manufactory/assembly/testing will be influenced? / 加工,装配,测试是否影响?

Feedback from manufacturing engineer? 生产制造装配是否影响，反馈?

测试线需要更改设定参数，1.功能测试PMB进气温度判断标准更改；2.充氮压力设置需要更改。

Influence on project&product delivery time? 对项目&产品交货时间的影响?

无

Influence on manufacturing cost / 对生产成本的影响?

☐ Increase/增加 ,金额____ ☐ Decrease/降低 ,金额____ ☑ No change/不变

Production/签名 Li hua

6. Influence on supplier part? Feedback from Supplier/Purchasing? ☑ NA.(新零件放行,不涉及库存)

Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier?
变更后的零件，生产可行性供应商是否确认？反馈?

Influence on purchasing part delivery time? 对采购件交货时间的影响?

涉及到变更的零件，博世已出PR，供应商还未来得及生产的订单，是否已经通知供应商取消. ☐ no/否 ☐ yes/是

7: Does the change influence on purchasing cost ? 变更对采购成本的影响?

☐ Increase/增加，金额____ ☐ Decrease/降低 ,金额____ ☑ no change/不变

PPS for purchase parts/签名 ____
(Below only for DCAC)
DCAC Mech. 签名 ____
DCAC软/硬件 签名 ____
Quality: 签名 ____

8. Changed parts Stock& treatment / 需要变更零件的库存统计和处理?
(只针对涉及到已经生产过的零件变更，新放行零件和本条不相关) ☐ 涉及到已经生产过的零件变更

RBCW仓库和RBCD仓库，是否有库存,	☐ 无库存	☑ 有库存，库存数量____允许混合发货____		COS /签名 Meng Fanlong
供应商处是否有库存,	☑ 无库存	☐ 有库存，库存数量____		
在途，运输中是否有库存	☑ 无库存	☐ 有库存，库存数量____		PPS /签名 ____
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☑ 可以继续使用完	☐ 改制后使用	☐ 报废	PPS for purchase parts/签名 ____ Quality for DCAC:/ 签名 ____
	☐ 改制费用____	☐ 报废成本____		

9. DFMA & Sample prodcution OPL? ☑ no/否 ☐ yes/是

DFMA 会议纪要，请附件说明；如果变更无需DFMA或前期PDECR已经做过DFMA，请说明并列出之前的PD_ECR号 Any OPL not closed from last time sample production? 上次样件生产是否仍有OPL没有完成? 请说明或者附件说明

10. PM alignment? (Influence on project Timeline? Cost?);
是否同意对项目时间和成本的影响____; 需求样品订单数量____;

PM/签名 俊龙 11.6.

11. 如果涉及以下情况需TCR3 审核确认。 不涉及则 ☑ N.A

☐ 库存报废,金额____; ☐ 采购成本增加,金额____; ☐ 生产成本增加,金额____

Step 3.2 Quality Assurance Items / 检查项目

Validations 验证	Plan finish date 计划完成日期	Resp.person 负责人	Comments 备注
☐ Capability Studies_ CMK			NA
☐ Capability Studies_ MSA			NA
☐ MAE release			NA
☐ Cleanness test			NA
☐ QZ test			NA
☐ 200h PDL			NA
☑ Test program or fixture etc.			功能台架进气温度范围需要更新

QMM/签名 Zou Lizhu

Step 3.3 Affected documents Check/ 影响文档检查

1 Interface FMEA-relevant / IFMEA:	☐ no/否 ☑ yes/是	Resp.person:	Fan Change	Due date 2025.11.07
2.Product FMEA-relevant / DFMEA:	☑ no/否 ☐ yes/是	Resp.person:		Due date: ____
3. Special Charecteristics relevant/PSC:	☑ no/否 ☐ yes/是	Resp.person:		Due date: ____
4. IMDS relevant:	☑ no/否 ☐ yes/是	Resp.person:		Due date: ____
5 Offer drawing relevant:	☐ no/否 ☑ yes/是	Resp.person:	Fan Change/Li Pingzheng/Wang Xiaolong	Due date 2025.11.07
6. TCD relevant:	☐ no/否 ☑ yes/是	Resp.person:	Qian Jiong/Sun Xudong	Due date 2025.11.15
7. Norm, WB, HF... relevant:	☐ no/否 ☑ yes/是	Resp.person:	Wu Zhaohui	Due date 2025.11.10

Step 4 Technical feasibility & Validation plan approval / 技术可行性&验证计划批准

Design team leader	Development SM	SMP-TMS	RBCW/TCR3
Signature / 签名: 2025.11.06	Signature / 签名: 2025.11.26	Signature / 签名:	Signature / 签名: 2025

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Edition 06
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Effective date:20250610 2

Confidential

DC No. / 开发更改编号: PDECR_0053
Date /日期: 20251105

Step 5 Documents release / 文档发布

1. Documents release (drawing, offer drawing, Bom, Spec.,) / 文档发布 (图纸， 客户图纸， …)

Development confirmation/ 研发确认: __________

Date /日期:

Step 6.1 Implementation check list / 导入清单

Departments	Y/N	Description	Responsible	Due date
Development	N	Change BOMs & Drawings & Documents in POE system		
	Y	Inform documents update (check work-on can met requirements)	All designe	2025.11.07
	Y	Update Offer drawing, TCD, D-FMEA	All designe	2025.11.07
	N	Norm, WB, HF...		
	N	MoC， IMDS.		
Manufacturing	N	Related (Production/Testing) equipement be ready on site		
	Y	Related (Production/Testing) program be ready	Wu Zhaohui	2025.11.10
	N	Related (Production/Testing) tooling / cutting / fixture etc. be ready		
	N	Old tooling / cutting / fixture disposal		
	N	Old materials disposal		
	N	Planner update the planning sheet		
	N	Update FMEA		
	N	Update CP/FC (Control Plan/Flow Chart)		
	Y	Update WI/PDS (Include attachments.)--》测试记录表	Wu Zhaohui	2025.11.10
	N	First batch Mark, Special Mark (Inside Package)		
	N	First batch Mark, Special Mark (Outside Package)		
	Y	Training	Wu Zhaohui	2025.11.10
COS	N	Confirm the storage of old parts and coordinate the introduction date for new part (RM)		
	N	Confirm the delivery date of old parts and first delivery of new parts (FG)		
	N	Check sample orders which affected:material order of CKD		
	N	Confirm production scheduling according to the alignment, any changes share the information to MOEx,MFE		
	N	Confirm the old stock/ do prioritize delivery and inventory handling		
	N	Inform the first delivery to PMO		
PPS (PPS->PMQ/PQA)	N	Check sample orders which affected: material order of purchasing		
	N	Inform internal related departments(COS,MFE,MOEx)with		
	N	Update incoming inspection plan		
QMM1	N	Update testing programe on testing equipment		
	N	Update inspection plan for CKD parts		
EIS4	Y	Distribute the Offer drawing, TCD to customer	Qian Jiong/	2025.11.07
LOP	N	Check 10 digit material order		
PMO	N	Check sample orders which affected: Customer order		
	N	Inform Customer the first delivery information		

Step 6.2 Implementation date / 执行日期

Planned implementation date:

变更计划执行日期______2025.11.07______________________________

充氮压力试验

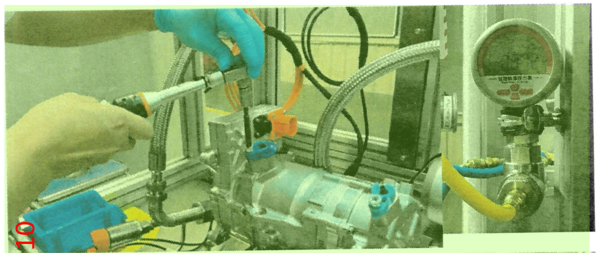
不同压力下保护帽弹开状态确认

重要备注：Offer drawing中要求压缩机拧松保护帽开启使用时需要能感受到内部正压力。

- **Background**：产线和质量部反馈当前设定压力过高，在拆保护帽时，保护帽冲击力较大。
- 基于不同压力下的试验结果，决定降低充氮压力：<u>0.15MPaA~0.25MPaA</u> → <u>0.13MPaA~0.19MPaA</u>。

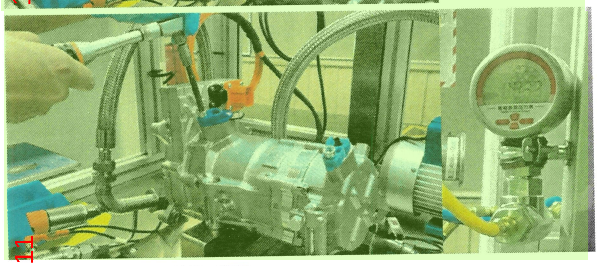
0.112MPaA

拆保护帽时无明显压力释放声，只在拔出时有“叭”声



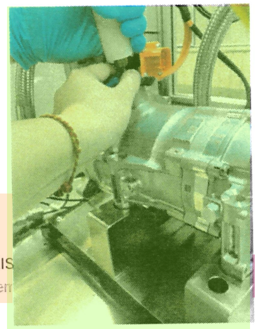
0.13MPaA

拆保护帽时能听到压力释放声，且冲击力很小



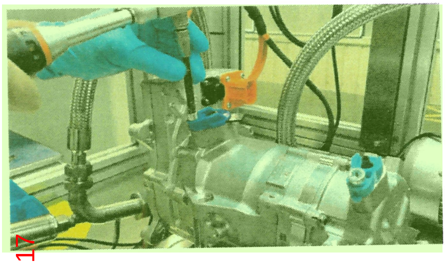
0.15MPaA

拆保护帽时能听到压力释放声，冲击力不大

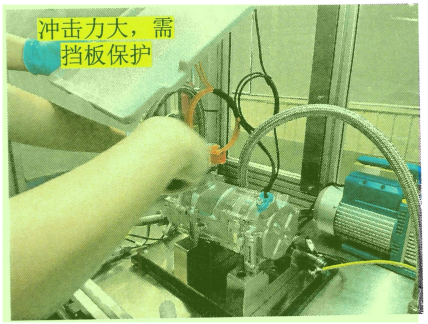


0.19MPaA

拆保护帽时能听到压力释放声，冲击力不大



0.25MPaA



结果：压力在0.13MPaA~0.19MPaA之间，能够在拧松保护帽过程中感受到压缩机内正压力，并且其冲击力也不大。

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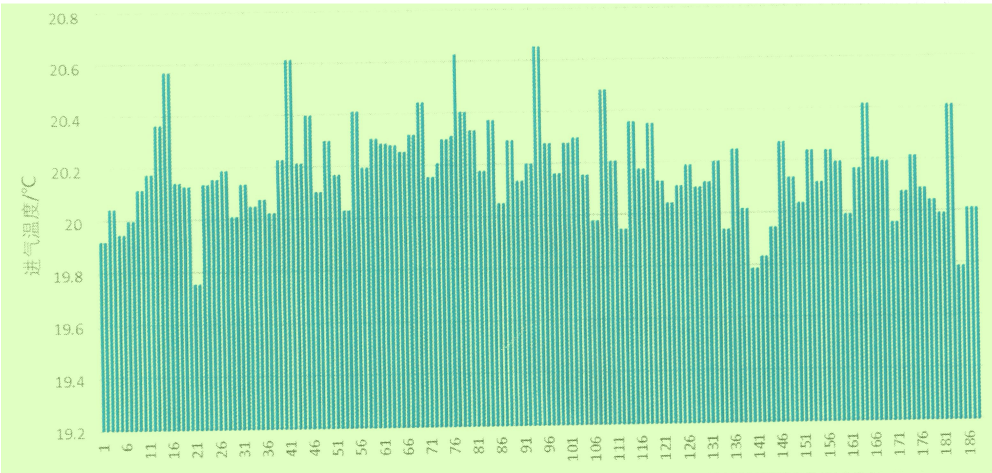
eCOMP气测台架进口温度

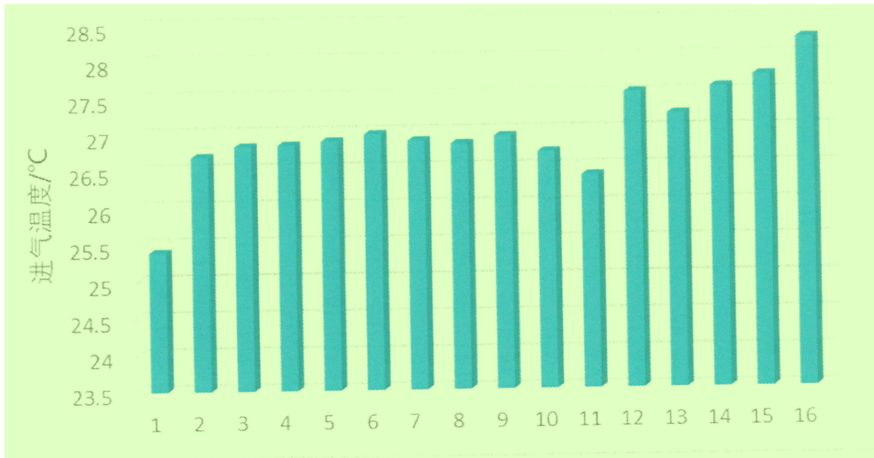
台架进口温度传感器记录数据

▶ 下线气测PMB中进气温度定义范围：**24.0±3℃**。由规范定义初期为数不多的几台压缩机数据确定了初始值。测试压缩空气由**FCM**集中供气，空气温度受季节影响较大，当前气测台架只有气体干燥作用，无控温作用。基于过去将近一年的实际生产数据，调整进气温度设定至**25.0±6℃**，对压缩机下线功能检查无影响。

12月份进气温度监控187组测试数据@57CC



八月份进气温度监控16台数据@45CC



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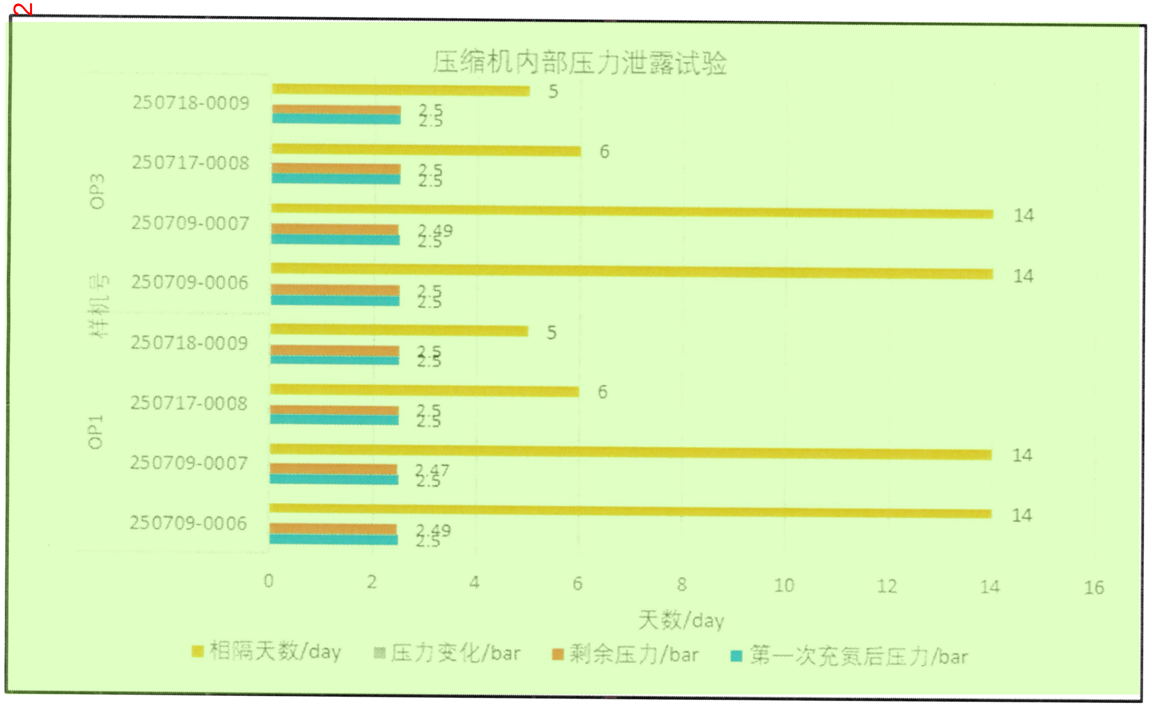
充氮压力试验

压缩机内部压力泄露试验



结果：压缩机内部2.5bar压力下半个月内几乎无泄露，说明当前保护帽装配状态下内部压力保压能力良好。

理论上若压力小于2.5bar，泄露率会更低。